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I shall be thanks to Allah Almighty who blessed us with own blessings.

We provide better service in the field of zoology.

SPECIAL THANX TO: ♡DR RABIA IQBAL ♡ DR SHUMAILA IQBAL ♡ DR FAHMIDA IQBAL ♡ PRO.DR MUSHTAQ HUSSAIN RAZVI ♡ SANAM SHAH SAHB ♡ M.I.A♡ DUA GO ALLAMA MUHAMMAD ALTAF HUSSAIN ATARI♡M.IQBAL♡

Q.no1 Define vector?

A vector is a living organism that transmits an infectious agent from an infected

animal to a human or another animal. Vectors are frequently arthropods, such as

mosquitoes, ticks, flies, fleas and lice

Q.no2 What are Soil dwelling insect?

Many other insects are also beneficial. •Soil-dwelling insects play important roles in

aeration, drainage, and turnover of soil, and they promote decay processes. •Other

insects serve important roles in food webs. •Soil-dwelling insects are hidden in the

soil and hard to detect until the sudden appearance of damage creates the demand for

rapid, curative treatment.

Qno3. What is Royal jelly?

Royal jelly is a honey bee secretion that is used in the nutrition of larvae, as well as

adult queens. It is secreted from the glands in the hypopharynx of nurse bees, and

fed to all larvae in the colony, regardless of sex or caste.

Qno4 Describe Gaseous exchange in Ophiuroids?

Ten bursa on the bottom of the central body disk. - Cilia within the sacs direct water

flow so that oxygen can be absorbed from the water - Water flows through cilia or

muscular contraction. - Oxygen is transported by the hemal system.

Q.no5 Class Ophiuroidea belong to which phylum?

Phylum Echinodermata

Q.no6 Paurometabolous or metamorphosis?

Paurometabolous is the process of gradual metamorphosis. Paurometabolous is the

type of growth that occurs in insects such as cockroaches, which includes just three

stages, i.e. egg, nymph, and adult. It is called incomplete metamorphosis or

paurometabolous since it lacks the pupa stage.

Q.No7 Classes of hormones in insects.

There are two main classes of hormones in insects:

1) The true hormones produced by epithelial glands and belonging to the

- ecdysteroids or juvenile hormones.
- 2) The neuropeptide hormones produced by neurosecretory cells.

Qno8 exchange in Ophiuroids?

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- hormones.
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Qno12. Explain Mating Behavior in Insects? long

- Mating behavior:
- Mating behavior is typically viewed as comprising all events from pair
- formation through courtship to the final breakup of the mating pair.
- Locating and recognizing mates:
- - Chemical attractants
- - Humming sound

- Courtship ritual.

- Mating:

- Internal fertilization by insertion of the male intromittent organ into the female genital
- tract for deposition of sperm is the usual method of copulation.
- BY PPT: Most insects have complex mating behaviors for locating and recognizing a potential
- mate, for positioning a mate for copulation, or for pacifying an aggressive mate.
- Mating behavior may involve pheromones (moths, order Lepidoptera), visual signals (fireflies,
- order Coleoptera), and auditory signals (cicadas, order Homoptera; and grasshoppers, crickets,
- and katydids, order Orthoptera).

- **Qno13 Hemal system in echinoderm?**

- in echinoderms, the blood vascular system is called hemal system, which is poorly developed in
- star fish and plays little role in circulation. The system consists of small fluid-filled sinus-
- channels that lack a distinct lining. The haemal channels are actually intercommunicating spaces
- and are not true blood vessels.

- **Qno14 Dermal Branchiae?**

- - External gill-like structures are scattered across their body surface.
- - Gas exchange.
- - Papulae.
- - Cilia beat and ventilate the respiratory surface.

- - Outer ectodermal epithelium, gland cells and a small basiepithelial nerve plexus.
- - Papulae function as a simple site of respiratory exchange.

Qno15 Aristotle's Lantern in Echinoids?

- - Conical shaped chewing apparatus projecting from mouth.
- - Complex arrangement of muscles and calcareous teeth constituting about 35 ossicles.
- - Teeth are extruded to scrape algae and other food from rocks.
- - Five strong teeth, 40 skeletal ossicles and 60 muscles.
- - Masticating the food.
- - Feed on algae, bryozoans, coral polyps, and dead animal remnants.

Q no16 Discuss Class Asteroidea?

- 1,900 species and second largest group in the Echinodermata.
- They often live on hard substrates in marine environments.
- - Thin folds of the body wall, called dermal branchiae.
- - Pincher-like structures called pedicellariae.
- - Ambulacral groove and paired rows of tube feet.
- - Suction disks sensory receptors are distributed over the surface of the body and tube
- feet.
- - Sea stars are dioecious, and external fertilization is the rule.

- Bipinnaria larva.

Q.no17 Aristotle's Lantern in Echinoids?

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- and tube feet.
- ▪ Sea stars are dioecious, and external fertilization is the rule.
- ▪ Bipinnaria larva
- **Q.no19 What is trapping?**
- Insect traps are used to monitor or directly reduce populations of insects or other
- arthropods, by trapping individuals and killing them.

- **Q.no20 Discuss Stomach of Star fishes?**

- A sea star has 2 stomachs, the cardiac stomach and the pyloric stomach. It can push
- the cardiac stomach out of its mouth, in the center of its underside, to engulf prey or
- insert it into prey (between 2 shells, for example). The stomach then secretes a
- powerful digestive enzyme to break down the prey.

- **Q.no21 Mating Behavior in Insects? Long**

- Mating behavior: Mating behavior is typically viewed as comprising all events from
- pair formation through courtship to the final breakup of the mating pair. Locating
- and recognizing mates: - Chemical attractants - Humming sound - Courtship ritual
- mating: Internal fertilization by insertion of the male intromittent organ into the
- female genital tract for deposition of sperm is the usual method of copulation. BY PPT: Most insects have complex mating behaviors for locating and recognizing a
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a simple site of respiratory exchange.

- **Q.no24 Skeleton of sea urchin? Explain**
- Sea urchins are rounded, and their oral end is oriented toward the substrate.
- •Their skeleton, called a test, consists of 10 closely fitting plates that arch between
- oral and amoral ends. Five rows of ambulacral plates have openings for tube feet,

- and alternate with five inter ambulacral plates, which have tubercles for the
- articulation of spines. The base of each spine is a concave socket, and muscles at its
- base move the spine.
- •Spines are often sharp and sometimes hollow, and they may contain venom
- dangerous to swimmers.

- **Q.no25 What is Garden centipedes?**

- Members of the class Symphyla (sim-filah) (Gr. sym, same phyllos, leaf) are small
- arthropods (2 to 10 mm in length) that occupy soil and leaf mold, superficially
- resemble centipedes, and are often called garden centipedes.

- **Q.no26 Phylum of sea urchin?**

- Sea urchin is invertebrates and belong to phylum Echinoderm.

- **Q.no27 Explain Circulatory system?**

- The circulatory system of insects is similar to that described for other arthropods,
- although the blood vessels are less well developed. Insects, like all other arthropods,
- have an open circulatory system which differs in both structure and function from
- the closed circulatory system found in humans and other vertebrates. In a closed

- system, blood is always contained within vessels (arteries, veins, capillaries, or the
- heart itself).

- **Q.no28 Asexual process in chilophores?**

- Although fragmentation, fission, and budding are methods of asexual reproduction
- in a number of fungi, the majority reproduce asexually by the formation of spores.
- Spores that are produced asexually are often termed mitospores, and such spores are
- produced in a variety of ways.
- .Name of animal of diplopoda?
- Members of the class diplopods are Millipedes
- Most millipedes are slow-moving detritivores, eating decaying leaves and other dead plant
- matter

- **Qno 29. What is Trepang ?**

- The body wall of sea cucumbers, when boiled and dried, is known as trepang in Asian countries.

- **Qno30. Describe Starfish stomach?**

- The mouth opens to a short esophagus and then to a large stomach that fills most of the coelom
- of the central disk.
- The stomach is divided into two regions.

- •The larger, oral stomach, sometimes called the cardiac stomach, receives ingested food.
- •It joins the smaller, aboral stomach, sometimes called the pyloric stomach.

Qno31 Discuss Complete metamorphosis?

- Complete metamorphosis is the type of insect development that includes egg, larva, pupal, and
- adult stages, which differ greatly in morphology. The lifecycle of butterflies, ants, fleas, bees,
- beetles, moths, and wasps are examples of the complete metamorphosis.

Qno32 Explain Ventilating process in insects? long

- Most insects have ventilating mechanisms that move air into and out of the tracheal system.
- • Insects do not breathe through their mouths as we do.
- • The do not have lungs and their blood, which is a watery, yellowish liquid, does not
- carry oxygen and carbon dioxide around their bodies.
- • Insects have a system of tubes, called tracheae, instead of lungs.
- • These tracheae penetrate right through the insect's body.
- • Air enters the tracheae by pores called spiracles.
- • These spiracles are found on each side of the insect's abdomen. Each segment of the
- abdomen has a pair of spiracles.
- •As oxygen diffuses from the tracheae to the body tissues, and is not replaced by carbon dioxide,

a vacuum is created that draws more air into the spiracles. This process is called passive suction.

Q.no33 Explain Insects pollination? Long

- Insects are responsible for the pollination of approximately 65% of all plant species.
- •Insects and flowering plants have co evolutionary relationships that directly benefit
- humans. The annual value of insect-pollinated crops is estimated at $19 billion per
- year in the United States. Insect pollination is crucial to most gardens and is as simple
- as insects like bees, butterflies and wasps flying from flower to flower in order to
- collect nectar.
- •In the process, pollen collects on their bodies and rubs off on other flowers that they
- visit.

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Qno39 what is Juvenile hormone?

- There are two main classes of hormones in insects:
- • the true hormones produced by epithelial glands and belonging to the ecdysteroids or juvenile
- hormones.

Qno40 explain Sea lily and star?

- PHYSICAL CHARACTERISTICS:
- Sea lilies have a stalk below the crown and look like flowers. Feather stars have a cluster of
- tentacles below the crown and look like ferns.
- Feather star are white, purple, brown, green. Or violet. Sea lily are white.
- HABITAT:

- Most sea lilies live in deep water and most feather star live on coral reefs.
- DIET:
- Both are planktonic
- **Qno41 Locomotion in Sea Cucumber? Long repeated**
- Sea cucumbers are mostly sluggish burrowers and creepers, although some swim by undulating

  their bodies from side to side.
- •Locomotion using tube feet is inefficient, because the tube feet are not anchored by body wall
- ossicles.
- ‘’Locomotion more commonly results from contractions of body-wall muscles that produce
- wormlike, locomotor waves that pass along the length of the body.
- One way that sea cucumbers can confuse or harm predators is by propelling their own toxic
- internal organs from their bodies in the direction of an attacker.

  •The organs grow back, and it may save them from being eaten.
-
- **Q.no42 Do insects have memory?**
- Studies over the past century have discovered that many insects, like humans,
- acquire more than one type of spatial memory, that they acquire these memories at
- different rates and that, as they become more familiar with an environment, they

- change which memories they use

- **Q.no43 Sea lily and star?**

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- their own toxic internal organs from their bodies in the direction of an attacker. •The
- organs grow back, and it may save them from being eaten.
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- **Q.no45 Synchronous Flight?**
- • Direct or synchronous flight is a mode of transportation that is fueled by wing
- muscles that insert directly into the wing base.
- • The insertion point of the wing is hinged which enables the muscles downward
- movements to lift the wing portion upward and upward movements pull the wing
- portion downward.

- **Q.no46 How cycliophores produce by asexually?**
- •It can reproduce asexually by producing a larva stage, called the Pandora that is
- released and settles on the same host.
- •Because of this asexual reproduction, large number of symbion can build up on one
- host.

- **Q.no47 Nervous System of Insects? Long repeated ?**
- The central nervous system consists of a series of ganglia that supply nerves to
- successive segments of the body. - The three main ganglia in the head

- (protocerebrum, deutocerebrum, and tritocerebrum) commonly are fused to form the
- brain. - The rest of the ganglionic chain lies below the alimentary canal against the
- ventral body surface. - - In most insects the number of separate ganglia has been
- reduced by fusion. The last abdominal ganglion always serves several segments. - In
- homopterans and heteropterans all the abdominal ganglia usually fuse with
- mesothoracic and metathoracic ganglia.

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- mesothoracic and metathoracic ganglia

- **Qno50 Class Crinoidea: General Features?**

- - Stem and a crown consisting of a cup-like central body known as the theca.
- - Five rays or arms.
- - Mouth surrounded by feeding arms and is linked to a u-shaped gut.
- - Passive suspension feeders.
- - Dioecious
- - Gonads are located in the pinnules but few in the arms.

- **Qno.51 What are the Ossicles?**

- Series of overlapping skeletal plates known as ossicles.
- The echinoderm skeleton consists of a series of calcium carbonate plates called ossicles.

- **Qno52 What is Alarm pheromones?**

- Released to warn other individuals of danger; may cause orientation

- toward the pheromone source and elicit a subsequent attack or flight from the source.

- **Qno53 insect have memory?**
- Research has demonstrated that insects are capable of some learning and have a memory.
- For example, bees (order Hymenoptera) instinctively recognize flowerlike objects by their shape
- and ability to absorb ultraviolet light, which makes the center of the flower appear dark.

- **Qno54 Sea star regeneration?**
- Sea stars are well known for their powers of regeneration.
- •They can regenerate any part of a broken arm.
- •In a few species, an entire sea star can be regenerated from a broken arm if the arm contains a
- portion of the central disk.
- Regeneration is a slow process, taking up to a year for complete regeneration.

- **Qno.55 for stages of metamorphosis?**

Egg

- larva
- pupa
- adult

- **Qno56 What are pheromones?**
- Pheromones are chemicals an animal releases that cause behavioral or physiological changes in

- another member of the same species.
- •Zoologists have described many different insect uses of pheromones.

Qno57 Define vector?

- A vector is a living organism that transmits an infectious agent from an infected animal to a
- human or another animal. Vectors are frequently arthropods, such as mosquitoes, ticks, flies,
- fleas and lice

Qno58 soil dwelling insect important?

- Many other insects are also beneficial.
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- promote decay processes.
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- damage creates the demand for rapid, curative treatment.

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Q.no61 Four stages of metamorphosis?

• Egg

• Larva

• pupa and

• Adult.

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changes in another member of the same species. •Zoologists have described many

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Qno63 What are centrepeds?

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small arthropods (2 to 10 mm in length) that occupy soil and leaf mold,

- superficially resemble centipedes, and are often

  Called garden centipedes.

Qno64 Give examples of Acoelomate?

- Acoelomate Examples
-

Others are parasitic and often pathogenic organisms that live within other animal organisms. Examples of flatworms include planarians, flukes, and tapeworms. Ribbon worms of the phylum Nemertea have historically been considered to be acoelomates.

Qno65 Describe body types of sponges?

Body types: Sponges have three body types: asconoid, syconoid, and leuconoid. Asconoid sponges are tubular with a central shaft called the spongocoel. The beating of choanocyte flagella force water into the spongocoel through pores in the body wall.

Qno66 what are Malpighian tubules?

Malpighian tubules are slender tubes normally found in the posterior regions of arthropod alimentary canals. Each tubule consists of a single layer of cells that is closed off at the distal end with the proximal end joining the alimentary canal at the junction between the midgut and hindgut.

Qno67 Circulatory system of archinids?

The circulatory system of the arachnids is an open system with hemolymph circulating in tissue sinuses. Special venous channels conduct hemolymph from the tissues to the heart, from which it is pumped through a series of blood vessels back to the sinuses.

Qno68 what is sea daisy?

- The class Concentricycloidea (ME consentrik, having a common center and Gr. kykloeides, like a
- circle) contains a single described species, known as the sea daisy.

• Sea daisies have been recently discovered on debris in deep oceans.

Qno69 what are tympanic organ?

- Tympanal (tympanic) organs are in the legs of crickets and katydids (order Orthoptera), in
- the abdomen of grasshoppers (order Orthoptera) and some moths (order Lepidoptera), and in the

thorax of other moths

Qno70 What is hydra and describe it's asexual reproduction?

- Hydra is a common freshwater hydrozoan that hangs from the underside of floating plants in clean
- streams and ponds.
- • Hydra lacks a medusa stage and reproduces both asexually by budding from the side of the polyp and

sexually.

Qno71 Sea star belong to which class of echinodermata?

The sea stars belong to the class **Asteroid.**

Qno72 Differentiate between bipinnaria larva and brachiolario larva?

Answer: brachiolaria is the second stage of larval development in many starfishes it follows the bipinnaria. Brachiolaria have bilateral symmetry, unlike the adult starfish, which have a pentaradial symmetry.

Qno73 Is fish contains pheromones?

Of course, alarm pheromones are not only found in fish; they are found everywhere also in plants etc.

Qno74 what is setae?

In biology, setae /ˈsiːtiː/ (singular seta /ˈsiːtə/; from the Latin word for "bristle") are any of a number of different bristle- or hair-like structures on living organisms.

Qno75 write the name of the subphylum of arthropods?

The phylum Arthropoda is commonly divided into four subphyla of extant forms: Chelicerata (arachnids), Crustacea (crustaceans), Hexapoda (insects and springtails), and Myriapoda (millipedes and centipedes)

Qno76 Describe the subphylum of sea cucumber ?

Sea cucumbers are echinoderms from the class Holothuroidea. They are marine animals with a leathery skin and an elongated body containing a single, branched gonad. Sea cucumbers are found on the sea floor world wide.

Qno77 What is fluke?

In parasitology, a fluke is a common name given to parasitic flatworms of the class Trematoda (trematodes).

Qno78 Differentiate between ectotherm and hetrotherm?

- ***Ectotherms***
- • Many insects are ectotherms (animals in which their heat source is primarily from the
- environment)
-

- *__Hetrotherms__*
- • Heterothermy ("hetero", different) occurs in animals which are able to switch from endothermy to ectotherm**y.**

Qno79 What is the sea star phylum name?

Sea star phylum name is

Phylum echinodermata.

Qno80 Describe polyphyletic origin?

- Polyphyletic origin .
- of these groups implies convergent evolution of remarkably similar
- Arthropodan features in all three (or at least two of three) phyla of Arthropods.
- The circulatory system of the arachnids.
- is an open system with hemolymph circulating in tissue sinuses. Special venous
- channels conduct hemolymph from the tissues to the heart, from which it is

pumped through a series of blood vessels back to the sinuses.

Qno81 Discuss bivalve internal fertilization?

- describe bivalve internal fertilization
- Sperm is drawn into a female's gills with the inhalant water and internal
- fertilization takes place. The eggs hatch into glochidia larvae that develop within
- the female's shell. Later they are released and attach themselves parasitically to

the gills or fins of a fish host.

Qno82 What is the scientific name of human?

- The scientific name
- of human is Homo
- sapiens.
- • When the genus is
- understood, the binomial
- name can be abbreviated

H. sapien

Qno83 Describe the characteristics of annelids?

Characteristics of annelids:

The Annelids are coelomate and triploblastic.

They exhibit organ system level organization.

Their body is segmented.

They respire through their body surface.

Nephridia are the excretory organs.

- **Qno84 What is Schizocoel hypothesis?**
- **Schizocoel hypothesis:**
- Coelom is formed by the splitting of mesoderm i.e. protostomes. It suggests that a triploblastic acoelomate (flatworm) body form is the ancestor of the coelomate.

Qno85 Define pinacocytes?

- Thin, flat cells, called pinacocytes, line the outer surface of a sponge. Pinacocytes may be mildly
- contractile and their contraction may change the shape of some sponges. In a number of sponges,
- some pinacocytes are specialized into tube like, contractile pinacocytes, which can regulate water
- circulation.

- **Qno86 What is Determinate cleavage?**

Determinate cleavage:

- Cleavage is a process of rapid mitotic cell divisions of the zygote to form number of cells for
- building up the offspring's body.It results in developmental fate of cells being set early in
- embryo development. Cell division in a fertilized or unfertilized egg resulting in daughter cells

that are no longer able to produce a complete embryo by themselves.

- **Qno87 Describe the body form of sponges?**
- Zoologist have described three sponge body forms:
- 1. Asconoid
- 2. Syconoid
- 3. Leuconoid
- 1=The simplest and least common sponge body form is the ascon. Simplest type of canal system
- found in asconoid sponges. Ascon sponges are vaselike. Ostia are the outer openings of

- porocytes and lead directly to a chamber called the spongocoal
- 2=In the sycon body form, the sponge wall appears folded.Water enters a sycon sponge through
- openings called dermal pores.
- 3=Leucon sponges have an extensively branched canal system. Water enters the sponge through

  ostia and moves through branched incurrents canals, which lead to Choanocyte-lined chambers.

- **Qno88 Define Ocelli?**
- Ocelli are concerned in some way with flight, and insects which are good at flying tend to have
- well developed ocelli. Ocelli consist of 500 to1,000 receptor cells beneath a single cuticular lens.
- Ocelli are sensitive to changes in light intensity. Ocelli are simple photoreceptors. They consist

  of a single lens and several sensory cells

Qno89 Describe the subphylum of sea cucumber?

The subphylum of sea cucumbers is subphylum Echinozoa**.**

Qno90 Describe the method of reproduction in hydra?

- **Reproduction mode**:
- The common asexual method of reproduction by hydras is budding.
- Buds originate at the junction of the stalk and gastric regions. The bud begins as a hemispherical
- outpouching that eventually elongates, becomes cylindrical, and develops tentacles. The bud

then pinches off and a new individual becomes independent.

Shab.e.taranum rahy ga ashkoon k ikhtasar tak

Iqbal

Phir bhe Khud ko tanha Pao to mujhy awaz Dena

Special Thanx to

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- **03460004097**
- تمہارے لیے نفرت کل بھی نہیں تھی تھے آج بھی نہیں
-
- دولت کے پجاری ہیں یہاں محبت کی لاج بھی نہیں
-
- اشک بھی ہیں میرے دامن میں پھر پھول ہیں مرجھائے کیوں کیایہاں ہمدردیوں کا رواج نہیں
-
- اک اجڑی سیج کی ٹوٹی آس ہوں جس کا کوئی سرتاج بھی نہیں
- کچھ ٹوٹے اس طرح سے کہ بکھر تے چلے گے
- میں تو وہ مریض عشق ہوں جس کا کوئی علاج بھی نہیں
-
- مجھے تو برباد کر گے ہجر ے یار کی لمبی مسافت
- پر تجھے کیا ہوا تیری وہ مسکان نہیں حسن پی وہ راج بھی نہیں
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- نہیں مانگی خوشیوں کی بھیک کسی سی آج تک
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- ♡♡اقبال♡♡
- مجھے اپنے غموں سے پیار ہے میری زندگی کسی کی خوشیوں کی محتاج نہیں

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- نہیں کوئی ضرورت ہمیں یاد رکھنے کی
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جہاں ذکر وفا ہو گا ہم خود یاد آئیں گے

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